

ORANGE COUNTY AP, VA, USA

WMO: 722167

Lat: 38.247N

Lon: 78.046W

Elev: 469

StdP: 14.45

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03718

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.7	(c) 18.3	(d) -0.2	(e) 5.5	(f) 20.8	(g) 4.5	(h) 7.1	(i) 25.3	(j) 20.5	(k) 47.5	(l) 18.7	(m) 43.5	(n) 2.1	(o) 310	(p) 0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.3	(c) 93.2	(d) 76.0	(e) 90.6	(f) 75.5	(g) 88.4	(h) 74.8	(i) 80.5	(j) 86.4	(k) 78.5	(l) 86.1	(m) 77.1	(n) 85.2	(o) 5.6	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	153.5	82.9	76.9	142.5	81.8	74.6	131.6	80.2	44.4	86.4	42.2	87.8	40.8	84.1	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.4	14.9	12.2	DB	5.3	97.7	5.4	2.7	1.4	99.6	-1.8	101.2	-4.8	102.8	-8.7	104.7	
			WB	4.5	81.5	5.3	2.0	0.7	82.9	-2.4	84.1	-5.4	85.2	-9.2	86.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	35.9	38.1	45.5	55.9	64.9	72.9	77.1	75.4	69.2	57.8	46.5	39.4	(6)
	DBStd	16.60	9.80	9.80	10.05	8.66	7.68	5.48	4.54	4.66	6.28	8.27	8.28	8.73	(7)
	HDD50	1584	450	350	214	37	1	0	0	0	0	23	162	347	(8)
	HDD65	4299	903	753	606	295	103	6	0	0	29	254	556	794	(9)
	CDD50	4010	11	18	75	214	462	687	839	787	576	264	58	19	(10)
	CDD65	1246	0	0	1	22	98	243	374	322	155	30	1	0	(11)
	CDH74	11524	0	1	49	336	960	2229	3538	2847	1294	256	14	0	(12)
	CDH80	4421	0	0	2	87	279	823	1554	1152	455	68	1	0	(13)
(14) Wind	WSAvg	3.9	4.6	4.8	5.2	5.2	3.9	3.4	2.9	2.6	3.0	3.6	3.9	3.8	(14)
(15) Precipitation	PrecAvg	43.0	2.9	2.9	3.5	3.1	3.9	4.2	4.6	3.8	4.2	3.5	3.7	3.0	(15)
	PrecMax	63.1	9.2	7.5	8.4	9.4	8.7	12.2	11.4	11.8	12.8	12.5	12.1	7.1	(16)
	PrecMin	30.6	0.3	0.4	0.3	0.7	0.8	0.6	1.0	0.5	0.4	0.0	0.5	0.3	(17)
	PrecStd	7.1	1.7	1.8	1.7	1.7	1.9	2.7	2.4	2.2	3.1	2.6	2.3	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.1	71.9	79.4	87.6	89.8	94.9	98.4	95.2	93.8	87.9	75.6	70.2	(19)
		MCWB	62.4	59.8	61.1	68.5	74.8	77.7	78.4	76.8	75.3	72.0	63.8	63.6	(20)
	2%	DB	62.8	64.5	73.3	82.0	86.3	90.6	94.7	91.2	89.9	81.2	70.2	63.8	(21)
		MCWB	55.4	54.4	58.2	64.4	71.9	75.8	76.9	76.0	73.9	68.3	60.0	59.0	(22)
	5%	DB	57.3	59.5	69.6	78.6	82.4	88.1	90.8	89.9	85.8	76.4	65.5	59.0	(23)
		MCWB	51.1	49.3	56.7	62.4	70.2	74.6	76.1	75.8	72.1	65.6	57.2	53.3	(24)
	10%	DB	51.8	54.3	63.5	72.5	80.5	85.7	88.3	86.5	81.3	72.2	62.2	54.0	(25)
		MCWB	46.3	47.1	53.7	60.3	68.9	72.9	75.5	74.5	70.2	64.2	54.9	49.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.7	61.5	64.8	70.9	77.2	81.8	82.8	82.5	81.9	74.1	67.1	66.4	(27)
		MCDB	65.4	67.5	73.4	78.5	86.4	85.7	87.8	86.9	86.0	82.3	73.1	67.8	(28)
	2%	WB	57.6	56.8	61.8	67.8	74.4	78.1	80.4	80.0	76.8	71.7	62.9	60.8	(29)
		MCDB	61.9	63.1	70.2	78.2	82.7	87.0	87.7	85.4	83.0	78.2	66.2	63.3	(30)
	5%	WB	51.8	52.4	58.8	64.9	72.4	76.4	78.5	78.0	74.4	69.6	59.7	54.6	(31)
		MCDB	55.8	57.5	66.2	74.5	80.2	85.4	87.0	85.1	81.5	73.1	64.7	57.4	(32)
	10%	WB	46.4	47.3	55.1	62.2	70.5	74.6	77.2	76.5	72.5	66.2	55.4	49.8	(33)
		MCDB	50.7	53.7	62.9	69.9	77.0	82.5	85.6	83.8	78.9	71.4	59.1	52.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.4	19.6	21.7	23.7	21.5	20.7	20.3	20.0	20.2	21.4	21.2	18.2	(35)
		MCDBR	26.3	27.9	29.7	30.2	25.6	24.2	23.9	23.6	25.9	25.9	27.9	23.8	(36)
		MCWBR	20.3	18.9	17.4	14.8	12.7	10.7	9.0	9.5	10.9	14.3	17.7	18.3	(37)
	5% WB	MCDBR	22.8	24.3	26.0	25.5	22.3	21.6	20.6	19.6	20.1	19.7	21.6	20.3	(38)
		MCWBR	19.6	18.7	17.4	14.3	12.1	11.0	10.0	10.0	10.7	11.9	16.4	17.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.297	0.314	0.349	0.392	0.435	0.482	0.503	0.507	0.412	0.353	0.319	0.302	(40)
	taud		2.512	2.449	2.398	2.304	2.205	2.114	2.081	2.047	2.299	2.461	2.512	2.531	(41)
	Ebn at Noon		283	291	289	281	270	256	249	245	265	273	271	272	(42)
	Edh at Noon		24	30	35	40	45	50	51	52	38	29	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		698	953	1266	1580	1767	1942	1868	1681	1385	1070	787	614	(44)
	RadStd		58	107	82	120	163	134	103	82	162	121	61	55	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (64 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+234	+140	

Nomenclature: See separate page